

Virgo Dataset Preprocessing Summary

Notebook

virgo-preprocessing.ipynb

Phase

Corpus Preprocessing & Tokenization Pipeline

Purpose

Builds the complete preprocessing pipeline for Virgo Base by cleaning raw documents, classifying them into subject categories, training the tokenizer and generating binary token datasets.

Primary Dataset

HuggingFaceFW/FineWeb-Edu (sample-10BT) streamed directly using the Hugging Face datasets library.

Subject Categories

General, Physics, Mathematics, Computer Science, Biology, Chemistry, Engineering, History, Economics, Social Science, Philosophy and Literature.

Dataset Distribution

General: 40%, Computer Science: 10%, Physics: 8%, Mathematics: 8%, Biology: 6%, Engineering: 6%, Chemistry: 5%, History: 5%, Economics: 4%, Social Science: 3%, Literature: 3%, Philosophy: 2%.

Token Budget

Total token budget: 2 Billion tokens allocated according to the predefined category ratios.

Preprocessing Pipeline

Stream documents → Clean and normalize Unicode → Remove invalid text → Classify by keywords → Save JSONL shards → Train tokenizer → Tokenize documents → Generate category-specific binary files → Save checkpoints.

Text Cleaning

Performs Unicode normalization (NFKC), whitespace normalization, line-ending normalization and invalid text filtering.

Tokenizer

Custom Byte-Level BPE tokenizer with a vocabulary size of 45,000 and special tokens (, , , ,).

Binary Dataset Generation

Stores token IDs as uint16 in category-specific .bin files using buffered writing for efficient large-scale dataset generation.

Checkpointing

Maintains dataset_checkpoint.json to resume preprocessing without restarting. Progress is periodically saved along with token counts.

Outputs

virgo_tokenizer.json, token_bins/, raw_categories/, tokenizer_training_data/, dataset_checkpoint.json and virgo_progress.zip.

Engineering Highlights

Streaming dataset processing, keyword-based subject classification, memory-efficient buffering, binary serialization, checkpoint recovery, tokenizer training and scalable preprocessing for billion-token datasets.

Summary

This notebook establishes the complete preprocessing workflow for Virgo Base, transforming raw educational documents into categorized, tokenized and checkpointed binary datasets ready for large-scale language model pretraining.